Roshan Balaji Mahashabde

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Summary

Engineering graduate finishing an MSc in Applied Artificial Intelligence at Warwick, with three research internships and four peer-reviewed publications. I build, deploy, and test machine learning systems across LLMs, computer vision, and data pipelines, working in Python and PyTorch. I turn research into working software that reaches real users, and I have the right to work in the UK.

Education

MSc Applied Artificial Intelligence | University of Warwick Sept. 2025 – Sept. 2026

- Modules: Artificial Intelligence & Deep Learning, Fundamentals of AI and Data Mining, Programming for Artificial Intelligence
- Dissertation: Machine-Learning Detection of GNSS Multipath: Generalisation, Interpretability, and the Limits of Measurement Exclusion for Reliable Navigation

BE Electronics and Communication Engineering | RV College of Engineering Dec. 2021 – Aug. 2025

- Awarded a 2:1 (UK equivalent)
- Modules: Signal Processing, Embedded Systems, Machine Learning
- Final-year project: Large Language Model Assistant for Data Acquisition and Control Systems (published at IEEE INSPECT 2025)

Work Experience

Research Intern | U R Rao Satellite Centre, ISRO | Bengaluru, India Apr. 2025 – June 2025

- Deployed a cloud-native LLM assistant using Hugging Face, Unsloth, Gradio, and Docker, adopted by an engineering team of 20+ staff
- Fine-tuned open-source models on domain documentation, working directly with users to improve answer relevance for 3 active R&D teams
- Replaced a manual search across 5+ legacy systems with a single containerised pipeline, giving engineers instant access to documentation

Project Engineering Intern | IISc × WIRIN, RVCE | Bengaluru, India June 2023 – June 2024

- Built a real-time 360-degree computer vision system using OpenCV and multi-camera stitching, removing blind spots for autonomous vehicle testing
- Integrated live video from 4 cameras into a perception pipeline, reducing detection errors caused by occlusion
- Delivered the system to the Centre of Excellence for Connected Autonomous Vehicles at RVCE and IISc, where it became the primary perception input for ongoing research

Research Intern | IEEE | Bengaluru, India Nov. 2023 – Apr. 2024

- Trained XGBoost and KNN classifiers on custom datasets, with results that contributed to 2 peer-reviewed publications
- Ran sensitivity analysis and simulations across 6+ configurations to find the optimal design variant, cutting manual evaluation time for the team

Projects

Digital Twin for Predictive Maintenance of a Marine Gas Turbine Mar. 2026 – Apr. 2026

- Created a web-based digital twin using Python, Flask, and XGBoost to simulate turbine telemetry and forecast engine degradation from live sensor readings in real time
- Benchmarked four regression models and selected XGBoost, reaching an R^2 of 0.99, then added SHAP explainability on an interactive fault-injection dashboard to surface the sensors driving each failure prediction

Predictive Healthcare Analytics: Diabetes Risk Modelling Jan. 2026 – Feb. 2026

- Built a diabetes-screening classifier on 229,781 CDC health records with RFECV feature selection and SMOTE, threshold-tuning an XGBoost model to 63% recall in flagging diabetes and prediabetes cases

- Hand-engineered composite risk features, such as a metabolic-syndrome index, then applied PCA and K-Means to segment patients into 4 clusters

NHS Prescribing Data Dashboard (Full-Stack) Oct. 2025 – Nov. 2025

- Shipped and tested a full-stack analytics dashboard using Python, Flask, SQLite, and Plotly.js to process and visualise NHS prescribing data across regions
- Processed 9.35M raw prescribing rows down to 1.88M using chunked Pandas ETL pipelines, then surfaced regional price variation across the cleaned dataset

Publications

- **Mahashabde, R.B.**, Chebbi, N.S., Practoor, P.P., Sharan, P. and Sharma, K. (2024) "Silver-TiO2 Nanoparticles Enabled Fiber Optic Sensor for Biosensing", 2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT). IEEE, pp. 1–7. doi: 10.1109/ICCCNT61001.2024.10725200.
- Chebbi, N., **Mahashabde, R.B.**, Nandi, S., Kumar, S., Jayanthi, P.N. and Sahana, B. (2025) "K-Nearest Neighbor Optimization of Silver-Graphene Fiber Optic Sensor for Lung Cancer Detection", Congress on Smart Computing Technologies (CSCT 2024). Smart Innovation, Systems and Technologies, vol. 121. Springer, Singapore. doi: 10.1007/978-981-96-6254-8_35.
- Chebbi, N.S., **Mahashabde, R.B.**, Shrinivas, B.M., Makam, P.K., Kumar, S. and Kiran, V. (2025) "Analysing Technology Ecosystem Business Models: A Predictive Modelling Approach", 2025 IEEE International Conference on Contemporary Computing and Communications (InC4). IEEE, pp. 1–7. doi: 10.1109/InC465408.2025.11256413.
- Chebbi, N.S., **Mahashabde, R.B.**, Usha Rani, K.R. and Barve, P. (2025) "The Chamber Assistant: A Language-Model Approach to Industrial Fault Correction and Documentation", 2025 IEEE International Conference on Intelligent Signal Processing and Effective Communication Technologies (INSPECT). IEEE, pp. 1–6. doi: 10.1109/INSPECT67393.2025.11350352.

Languages, Skills, and Certificates

Languages: English (Fluent)

Skills: Python, MATLAB, SQL, C/C++; PyTorch, TensorFlow, Scikit-learn, Hugging Face; OpenCV, Pandas, NumPy, XGBoost; HTML/CSS/JavaScript, Docker, Git, Flask, SQLite, Plotly.js, VS Code

Certifications: Foundation of Cloud IoT Edge ML, NPTEL (May 2024; ID NPTEL24CS26S552301938); Programming, Data Structures and Algorithms Using Python, NPTEL (Sept. 2023; ID NPTEL23CS95S14644024)